1 Divisibility and Factorization

Proposition 1.1. $a \mid b \ \& \ b \mid c \implies a \mid c$

Proposition 1.2. $c \mid a \ \& \ c \mid b \implies \forall m, n \in \mathbb{Z}, c \mid (ma + nb)$

Lemma 1.3. $\forall x \in \mathbb{R}, x - 1 < [x] \leq x$

Theorem 1.4 (Division Algorithm)**.** $\forall a, b \in \mathbb{Z}, b > 0. \exists$ *unique* $q, r \in \mathbb{Z}, a = bq + r \ \& \ 0 \leq r < b.$

Lemma 1.5. *Every integer greater than 1 has a prime divisor.*

Theorem 1.6. *There exists infinitely many primes.*

Proposition 1.7. *n is composite* $\implies \exists$ *at least one prime* $p \leq \sqrt{n}$ *dividing n.*

Proposition 1.8. *For* $n \in \mathbb{Z}^+, \exists$ *n consecutive composite numbers*

Conjecture (Goldbach)**.** *Every even integer* > 2 *is a sum of two primes.*

Definition (Mersenne prime)**.** *Suppose p is a prime. If* $2^p - 1$ *is a prime, it's a Mersenne prime*

Conjecture. $\exists$ *infinitely many Mersenne primes.*

Definition (Greatest Common Divisor)**.** *Let* $a, b \in \mathbb{Z}$*, then the greatest common divisor of a and b is the largest common divisors of a and b. Notation: (a,b)*

Proposition 1.10. $(\frac{a}{d}, \frac{b}{d} = 1) \iff (a, b) = d$

Proposition 1.11. *Let* $a, b \in \mathbb{Z}$*, not both 0. Then* $(a, b) = \min\{ma + nb > 0 : m, n \in \mathbb{Z}\} > 0$

Lemma 1.12. *Let* $a, b \in \mathbb{Z}^+, a \geq b$*. Let* $a = bq + r, q, r \in \mathbb{Z}$*. Then (a,b)=(b,r)*

Theorem 1.13. *For every* $y \geq 2, \sum_{p \leq y} \frac{1}{p} \geq \log\log y - 1$

Lemma 1.14. *Let p be prime and* $a, b \in \mathbb{Z}$*.* $p \mid ab \implies p \mid a$ *or* $p \mid b$

Corollary 1.15. *Let p be a prime,* $a_1, \ldots, a_n \in \mathbb{Z}, n \geq 2$*. If* $p \mid a_1 a_2 \ldots a_n$*, then* $p \mid a_j$ *for at least one j.*

Theorem 1.16 (Fundamental Theorem of Arithmetic)**.** *Every integer* $a > 1$ *can be represented uniquely as a product of primes:* $a = p_1^{a_1} p_2^{a_2} \ldots p_n^{a_n}$ *where* $p_i \neq p_j$ *if* $i \neq j$ *and* a_i *positive integers.*

Definition (Least Common Multiple)**.** *Let* $a, b \neq 0$*. The least common multiple of a and b, [a,b], is the least* $m > 0$ *so that* $a \mid m$*,* $b \mid m$*.*

Theorem 1.17. $(a, b)[a, b] = ab$

Theorem 1.18 (Dirichlet's Theorem on Primes in Arithmetic Progressions)**.** *If (a,b)=1, then* $\{an + b : n \geq 0\}$ *has infinitely-many primes.*

Lemma 1.23. *Let* $a, b \in \mathbb{Z}^+$ *and* $a, b \in \{4n + 1 : n \geq 0\}$*. Then* $ab \in \{4n + 1 : n \geq 0\}$

2 Congruences

Definition (Congruent)**.** *a is congruent to b modulo m, $m \in \mathbb{Z}^+$, if $m \mid (a-b)$*

Definition (Complete Residue System)**.** *Complete Residue System modulo m is a set, S, of integers such that every $n \in \mathbb{Z}$ is congruent to one and only one member of S*

Proposition 2.3. *$S = \{0, 1, \ldots, m-1\}$ is a complete residue system* $\pmod m$

Proposition 2.4. *Let $a, b, c, d \in \mathbb{Z}$. $a \equiv b \pmod m, c \equiv d \pmod m$. Then*

1. *$a + c = b + d \pmod m$*
2. *$ac \equiv bd \pmod m$*

Proposition 2.5. *$ca = cb \mod m \iff a \equiv b \pmod{\frac{m}{(c,m)}}$*

Definition (Reduced Residue System)**.** *A reduced residue system modulo m is a set of integers $r_1, \ldots, r_n$ such that if $(a, m) = 1$, then $a \equiv r_j \mod m$ for 1 and only 1 value of j.*

Definition (Euler $\phi-$ Function)**.** *$\phi(m)$ is the number of members in a reduced residue system mod m. $\phi(m) = \|\{y \in \mathbb{Z}, 1 \leq j \leq n, (j, n) = 1\}\|$*

Theorem 2.6. *Let $r_1, \ldots, r_n$ be a complete (reduced) system modulo m. Let $(a, m) = 1$. Then $ar_1, \ldots, ar_n$ is a complete (reduced) residue system modulo m.*

Theorem 2.7. *Let $ax \equiv b \pmod m$. Let $d = (a, m)$. If $d \nmid b$, then there are no solutions. If $d \mid b$, then there exists exactly d incongruent solutions modulo m.*

Definition (Multiplicative Inverse)**.** *A solution of $ax \equiv 1 \mod m$ is a (multiplicative) inverse of $a \pmod m$.*

Corollary 2.8. *The linear congruence $ax \equiv 1 \pmod m$ has a solution if $(a, m) = 1$ and the inverse is unique.*

Theorem 2.9 (Chinese Remainder Theorem)**.** *Let $m_1, \ldots, m_n > 0$ be positive integers and $(m_i, n_j) = 1, i \neq j$. Let $b_i, \ldots, b_n \in \mathbb{Z}$, then: $x \equiv b_1 \pmod{m_1}, \ldots, x \equiv b_n \pmod{m_n}$ has a unique solution $\pmod{m_1 \ldots m_n}$. Let $M = m_1 \ldots m_n$. $M_j = \frac{M}{m_j}, 1 \leq j \leq n$. $M_j x_j \equiv 1 \pmod{m_j}$.*

$$x = \sum_{j=1}^{n} b_j M_j x_j$$

Lemma 2.10. *Let p be a prime. The inverse of $a \pmod p$ is a iff $a \equiv \pm 1 \pmod p$*

Theorem 2.11 (Wilson's Theorem)**.** *If p is a prime, $(p-1)! \equiv -1 \pmod p$*

Proposition 2.12. *Let $n \in \mathbb{Z}^+, n > 1$. Suppose $(n-1)! \equiv -1 \pmod{n}$, then n is a prime.*

Definition (Wilson's Prime)**.** *A Wilson prime p is such that $(p-1)! \equiv -1 \pmod{p^2}$*

Theorem 2.13 (Fermat's Little Theorem)**.** *If p is a prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$*

Corollary 2.14. *Let p be a prime, and $p \nmid a$, then the inverse of $a \pmod{p}$ is a^{p-2}*

Corollary 2.15. *For all a and primes p, $a^p \equiv a \pmod{p}$*

Definition (Pseudoprime)**.** *If $2^n \equiv 2 \pmod{n}$, then n is a pseudoprime.*

Theorem 2.16 (Euler's Theorem)**.** *Let $\{a, m\} \in \mathbb{Z}, m > 0, (a, m) = 1$. Then $a^{\phi(m)} \equiv 1 \pmod{m}$*

3 Arithmetic Functions

Definition (Arithmetic functions)**.** *Arithmetic functions a(n) are functions whose domains are the positive integers*

Definition (Multiplicative)**.** *a(n) is multiplicative if whenever (m,n)=1, then a(mn)=a(m)a(n). If we can remove the restriction (m,n)=1 then the function is completely multiplicative.*

Definition (Additive)**.** *a(n) is additive if a(m+n)=a(m)+a(n)*

Theorem 3.1. *If f(n) is multiplicative, then $F(n) = \sum_{d|n} f(d)$ is also multiplicative.*

Theorem 3.2. *$\phi(mn) = \phi(m)\phi(n)$ whenever (m,n)=1*

Theorem 3.3. *$\phi(p^a) = p^a - p^{a-1}$ for every prime p and positive integer a.*

Theorem 3.4. *$\phi(n) = \prod_{i=1}^{k} p^a(1 - \frac{1}{p}) = n \prod_{p|n} (1 - \frac{1}{p})$, where p is prime*

Theorem 3.5. *$\forall n \in \mathbb{Z}^+, \Sigma_{d|n, d>0} \phi(d) = n$*

Definition. *Divisors of $n = \nu(n) = d(n) = \tau(n) = \Sigma_{d|n} 1$.*

Theorem 3.6. *d(n) is multiplicative*

Theorem 3.7. *$d(p^a) = a + 1$ for every prime p and positive integer a.*

Theorem 3.8. *Let $n = \Pi_{j=1}^{r} p_j^{\alpha_j}$. Then $d(n) = \Pi_{j=1}^{r} (\alpha_j + 1)$*

Definition. *$\sigma(n) = \Sigma_{d|n} d$*

Theorem 3.9. *$\sigma(n)$ is multiplicative*

Theorem 3.10. *Let p be a prime, $a \in \mathbb{Z}^+$. Then $\sigma(p^a) = \frac{p^{a+1}-1}{p-1}$.*

Theorem 3.11. *Let $n = \Pi_{j=1}^{r} p_j^{\alpha_j}$. Then $\sigma(n) = \Pi_{j=1}^{r} \frac{p_j^{\alpha_j+1}-1}{p_j-1}$*

Definition (Perfect)**.** *n is perfect if $\sigma(n) = 2n$, i.e., n is perfect if the sum of the proper divisors of n is n.*

Theorem 3.12. *n is an even perfect number $\iff$ $n = 2^{p-1}(2^p - 1)$, where $p, 2^p - 1$ are prime.*

Definition (Mobius Function)**.**

$$\mu(n) = \begin{cases} 1, & \text{if } n = 1 \\ 0, & \text{if } p^2 \mid n \text{ for some prime } p \\ (-1)^r & \text{if } n = p_1 p_2 \dots p_r \end{cases}$$

Theorem 3.13. *$\mu(n)$ is multiplicative.*

Lemma 3.14. *For $n \in \mathbb{Z}^+$, $\Sigma_{d|n}\mu(d) = \begin{cases} 1, & \text{if } n = 1 \\ 0, & \text{if } n > 1 \end{cases}$*

Theorem 3.15 (Mobius Inversion Formula)**.** *$f(n) = \Sigma_{d|n} g(d) \iff g(n) = \Sigma_{d|n}\mu(d) f(\frac{n}{d}) = \Sigma_{d|n}\mu(\frac{n}{d}) f(d)$*

4 Quadratic Residues

Definition (Quadratic Congruence)**.** *a is a quadratic residue modulo m if $x^2 \equiv a \pmod{m}$ has a solution and a is a quadratic nonresidue of m otherwise.*

Theorem 4.1. *Let p be an odd prime. Then $x^2 \equiv a \pmod{p}, p \nmid a$ has either 2 incongruent solutions or 0 solutions.*

Theorem 4.3. *Let p be an odd prime. $\exists \frac{p-1}{2}$ quadratic residues mod p and quadratic nonresidues mod p.*

Definition (Legendre Symbol)**.**

$$(\frac{a}{p}) = \begin{cases} 1, \text{if } a \text{ is a quad. res. mod } p \\ -1, \text{if } a \text{ is a quad. nonres. mod } p \end{cases}$$

Theorem 4.4 (Euler's Criterion)**.** *Let p be odd prime, $p \nmid a$. Then $(\frac{a}{p}) = a^{(p-1)/2} \pmod{p}$*

Proposition 4.5.

$$(\frac{a}{p})(\frac{b}{p}) = (\frac{ab}{p}), a \equiv b \pmod{p} \implies (\frac{a}{p}) = (\frac{b}{p}), (\frac{a^2}{p}) = 1$$

Theorem 4.6.

$$(\frac{-1}{p}) = (-1)^{(p-1)/2} = \begin{cases} 1, p \equiv 1 \pmod 4 \\ -1, p \equiv 3 \pmod 4 \end{cases}$$

Lemma 4.7 (Gauss)**.** *Let p be an odd prime, $p \nmid a$. Let n = no. of least positive residues of $a, 2a, \ldots, \frac{p-1}{2}a$ that are $> \frac{p}{2}$. Then $\frac{a}{p} = (-1)^n$*

Theorem 4.8.

$$(\frac{2}{p}) = (-1)^{(p^2-1)/8} = \begin{cases} 1, p \equiv 1, 7 \pmod 8 \\ -1, p \equiv 3, 5 \pmod 8 \end{cases}$$

Theorem 4.9 (Law of Quadratic Reciprocity)**.** *Let p,q be distinct odd primes.*

$$(\frac{p}{q})(\frac{q}{p}) = (-1)^{\frac{(p-1)}{2}\frac{(q-1)}{2}} = \begin{cases} 1, p \text{ and/or } q \equiv 1 \pmod 4 \\ -1, p, q \equiv 3 \pmod 4 \end{cases}$$

Lemma 4.10. *Let $N = \Sigma_{j=1}^{(p-1)/2}[\frac{ja}{p}]$ where (a,2p)=1. Then $(\frac{a}{p}) = (-1)^N$*

Definition (Legendre-Jacobi Symbol)**.** *P and Q are integers, Q ¿ 0 and odd, $P \neq 0$. $Q = q_1 q_2 \ldots q_s$. The Legendre-Jacobi Symbol $(\frac{P}{Q}) = \Pi_{j=1}^{s}(\frac{P}{q_j})$.*

Theorem 4.11. *Let $P, P' \in \mathbb{Z}, P \neq 0, Q, Q'$ are odd and positive. Then*

$$1.(\frac{P}{Q})(\frac{P}{Q'}) = (\frac{P}{QQ'}), 2.(\frac{P}{Q})(\frac{P'}{Q}) = (\frac{PP'}{Q}), 3.(\frac{P^2}{Q}) = (\frac{P}{Q^2}) = 1$$

$$4.(PP', QQ') = 1 \implies (\frac{P'P^2}{Q'Q^2}) = (\frac{P'}{Q'}), 5.P \equiv P' \pmod Q \implies (\frac{P'}{Q}) = (\frac{P}{Q})$$

Theorem 4.12. *Q odd, > 0. $(\frac{-1}{Q}) = (-1)^{(Q-1)/2}$, $(\frac{2}{Q}) = (-1)^{(Q^2-1)/8}$.*

Theorem 4.13 (Law of Quadratic Reciprocity)**.** *Let $P, Q > 0$, odd. Then $(\frac{P}{Q})(\frac{Q}{P}) = (-1)^{\frac{(P-1)}{2}\frac{(Q-1)}{2}}, (P, Q) = 1$*

5 Primitive Roots

Definition (Order)**.** *Order of a $\pmod m$ is the smallest positive integer n such that $a^n \equiv 1 \pmod m$. $ord_m a \leq \phi(m)$.*

Proposition 5.1. *Let $(a, m) = 1, m > 0$. If $a^n \equiv 1 \pmod m$, then $ord_m a \mid n$. So $ord_m a \mid \phi(m)$. Converse is also true.*

Proposition 5.2. *Let $(a, m) = 1, m > 0$. Let $i, j \geq 0, i, j \in \mathbb{Z}$. $a^i \equiv a^j \pmod m \iff i \equiv j \pmod{ord_m a}$.*

Definition (Primitive Root)**.** *r is a primitive root mod m if $ord_m r = \phi(m)$. Then (r,m) = 1. r is an imprimitive root mod m if $ord_m r < \phi(m)$.*

Proposition 5.3. *If r is a primitive root, then a set of reduced residues mod m is $r, r^2, \ldots, r^{\phi(m)}$*

Proposition 5.4. *Let $(a, m) = 1, m > 0$. If $n \in \mathbb{Z}^+, ord_m a^n = \frac{ord_m a}{(ord_m a, n)}$*

Corollary 5.5. *Let $(a, m) = 1, m > 0, n \in \mathbb{Z}^+$. Then $ord_m a^n = ord_m a \iff (ord_m a, n) = 1$.*

Corollary 5.6. *If a primitive root r mod m exists, then $\exists$ exactly $\phi(\phi(m))$ primitive roots mod m.*

Theorem 5.7 (Lagrange's Theorem)**.** *Let p be a prime. Let $f(x) = \sum_{j=0}^{n} a_j \cdot x^j, a_j \in \mathbb{Z}, p \nmid a_n$. Then $f(x) \equiv 0 \pmod{p}$ has at most n incongruent solutions mod p.*

Proposition 5.8. *Let p be a prime, $d \mid (p-1)$. Then $x^d - 1 \equiv 0 \pmod{p}$ has exactly d incongruent solutions mod p.*

Theorem 5.9. *Let p be a prime, $d \in \mathbb{Z}^+, d \mid (p-1)$. Then $\exists$ exactly $\phi(d)$ incongruent integers having order d mod p.*

Corollary 5.10. *Let p be prime. Then $\exists$ exactly $\phi(p-1)$ incongruent primitive roots mod p.*

6 Diophantine Equations

Theorem 6.2. *Consider ax+by=c, where $a, b, c \in \mathbb{Z}, d = (a, b)$. If $d \nmid c$, then there does not exist any solutions. If $d|c$, then there are infinite solutions. Let (x_0, y_0) be a solution, then all solutions are given by $x = x_0 + \frac{b}{d}n, y = y_0 - \frac{a}{d}n$*

A nonlinear diophantine can be proved to be not solvable by taking a modulo.

Definition (Primitive Pythagorean Triple)**.** *$x^2+y^2 = z^2, (x, y, z)$ is a primitive Pythagorean triple if $(x, y, z) = 1$. One of x or y is even.*

Theorem 6.3. *There are infinitely many primitive Pythagorean triples with y even and they are given by:*

$$x = m^2 - n^2, y = 2mn, z = m^2 + n^2$$

where $(m, n) = 1$ and exactly one of them is even.

Theorem 6.4 (Fermat)**.** *The Diophantine equation $x^4+y^4 = z^2$ has no nonzero integer solutions.*

Proposition 6.5. *If n_1, n_2 are expressible as the sums of two squares of integers, then $n_1 n_2$ is expressible as the sum of two squares of integers.*

Proposition 6.6. *Let p be prime, $p \equiv 1 \pmod 4$, then there exists $x, y \in \mathbb{Z}, k \in \mathbb{Z}^+$ such that $x^2 + y^2 = kp$.*

Proposition 6.7. *Every prime $p \equiv 1 \pmod 4$ is expressible as a sum of 2 squares. Also $p = 2$ is so expressible. (i.e. every prime $\not\equiv 3 \pmod 4$)*

Theorem 6.8 (Primes as the Sum of Squares)**.** *$n = a^2 + b^2 \iff$ each prime in the factorization of n that is $\equiv 3 \pmod 4$ has even exponent.*

Theorem 6.9. *$N = 4^m(8n + 7)$ cannot be represented as a sum of 3 squares*

Proposition 6.10. *Let n_1, n_2 be expressible as a sum of 4 squares. Then $n_1 n_2$ can be represented as a sum of 4 squares.*

Lemma 6.11. *Let p be odd prime, then $\exists x, y, k \in \mathbb{Z}, 0 < k < p$ such that $x^2 + y^2 + 1 = kp$*

Theorem 6.12. *If p is a prime, then $p = w^2 + x^2 + y^2 + z^2$*

Theorem 6.13 (Lagrange)**.** *All positive integers are expressible as the sums of four squares of integers*

Theorem 6.14 (Hilbert)**.** *$g(k)$ is the minimum k such that every integer can be expressed as the sum of $g(k)$ K powers, and it exists. $g(k) = 2^k + [\frac{3}{2}^k] - 2$ provided that $2^k\{(\frac{3}{2})^k\} + [\frac{3}{2}^k] \leq 2^k$ except for a finite number of cases, $\{x\} = x - [x]$*

7 Characters and Gauss Sums

Definition (Character)**.** *A character $\chi(n)$ has period p and is multiplicative, values are roots of unity, $\chi(n) = 0$ if $(n, p) > 1$.*

Definition (Trivial Character)**.** *$\chi(n) = 1$ for all $p \nmid n$ is a trivial character.*

Proposition 7.1.

$$\sum_{n=1}^{p} \chi(n) = \begin{cases} p - 1, \text{ if } \chi \text{ is trivial} \\ 0, \text{ otherwise} \end{cases}$$

Definition (Gauss Sum)**.** $G(\alpha, \chi) = \sum_{n \pmod p} \chi(n) e^{2\pi i n\alpha/p}, G(1, \chi) = G(\chi)$

Theorem 7.2. *$G(\alpha, \chi) = \chi(\alpha^{-1})G(\chi) = \bar{\chi}(\alpha)G(\chi)$, if $\alpha \neq 0 =, \alpha \neq np$*

Theorem 7.3. *$G(\alpha, \chi)G(\alpha, \bar{\chi}) = \chi(-1)p$, where χ is nontrivial and $\alpha \not\equiv 0 \pmod p$.*

Proposition 7.4. $\overline{G(\alpha, \chi)} = \chi(-1)G(\alpha, \bar{\chi})$

Corollary 7.5. $|G(\alpha, \chi)| = \sqrt{p}$

Definition. $g(\beta) = \sum_{n \pmod p} e^{2\pi i n^2 \beta/p} (= G(\beta, \chi)$ *for the Legendre symbol)*

Proposition 7.6. *For Legendre symbol,* $G(1, \chi) = \begin{cases} \sqrt{p}, \text{ if } p \equiv 1 \pmod 4 \\ i\sqrt{p}, \text{ if } p \equiv 3 \pmod 4 \end{cases}$

8 Continued Fractions

Definition. $[a_0, a_1, \ldots, a_n]$ *is a simple continued fraction if* $a_1, \ldots, a_n$ *are positive integers and* $a_0 \in \mathbb{Z}$

Proposition 8.5. *Let* $\alpha \in \mathbb{R}$. *Then* $\alpha \in \mathbb{Q} \iff \alpha$ *can be expressed as a finite simple continued fraction*

Definition. $C_i = [a_0, a_1, \ldots, a_i], 0 \leq i \leq n$ *is the ith convergent of* $[a_0, a_1, \ldots, a_n]$

Proposition 8.6. *Define* $p_0 = a_0, p_1 = a_0a_1 + 1, p_j = a_jp_{j-1} + p_{j-2}$ *for* $j \geq 2$ *and* $q_0 = 1, q_1 = a_1, q_j = a_jq_{j-1} + q_{j-2}$ *for* $2 \leq j \leq n$. *Then* $C_j = \frac{p_j}{q_j}$.

Proposition 8.7. $p_jq_{j-1} - p_{j-1}q_j = (-1)^{j-1}, 1 \leq j \leq n$.

Corollary 8.8. $(p_j, q_j) = 1, 0 \leq j \leq n$.

Corollary 8.9. $C_j - C_{j-1} = \frac{(-1)^{j-1}}{q_jq_{j-1}}, 1 \leq j \leq n$. $C_j - C_{j-2} = \frac{(-1)^j a_j}{q_jq_{j-2}}, 2 \leq i \leq n$.

Proposition 8.10. $C_0 < C_2 < C_4 \ldots C_5 < C_3 < C_1$

Lemma 8.11. $q_i \geq j, j \geq 0$

Proposition 8.12. *Let* $a_0, a_1, a_2, \ldots$ *be an infinite sequence of positive integers except for* a_0. *Then* $lim_{j\to\infty}C_j$ *exists.*

Proposition 8.13. *Let* $\alpha \in \mathbb{R}$. *Then* $\alpha \in \mathbb{R}/\mathbb{Q} \iff \alpha$ *can be expressed as a infinite simple continued fraction*

Proposition 8.14. *Expression of irrational* α *as infinite simple continued fraction is unique.*

Theorem 8.15. *Let* $\alpha \in \mathbb{R}/\mathbb{Q}$. *Let* $\frac{p_n}{q_n}$ *be the* n^{th} *convergent to* α. *Let* $a, b \in \mathbb{Z}, 0 < b < q_{n+1}$. *Then* $|q_n\alpha - p_n| \leq |b\alpha - a|$.

Corollary 8.16. *Let* $\alpha \in \mathbb{R}/\mathbb{Q}$. *Let* $\frac{p_j}{q_j}$ *be the* j^{th} *convergent of a simple continued fraction of* α. *Let* $a, b \in \mathbb{Z}, 1 < b < q_j$. *Then* $|\alpha - \frac{p_j}{q_j}| \leq |\alpha - \frac{a}{b}|$.

Definition. $[a_0, a_1, a_2]$ *is eventually periodic if* $\exists$ *positive integers N, P such that* $a_n = a_{n+p}, n \geq N$.

Theorem 8.17. *Let* $\alpha = \frac{P_0+\sqrt{d}}{Q_0}, P_0, Q_0, d \in \mathbb{Z}, d > 0, Q_0 \neq 0, d$ *not perfect square, and* $Q_0|(d - P_0^2)$. *Let* $\alpha_j = \frac{P_j+\sqrt{d}}{Q_j}, a_j = [\alpha_j], P_{j+1} = a_jQ_j - P_j, Q_{j+1} = \frac{d-P_{j+1}^2}{Q_j}.\alpha = [a_0, a_1, a_2, \ldots]$